A Truly Beautiful Mind

The big idea

This biographical piece traces the life of **Albert Einstein**: a slow-talking child who became the greatest physicist of the twentieth century, and a scientist who later worked tirelessly for **world peace**. "Beautiful mind" refers to both his genius and his humanity.

Life in brief

Year	Event
1879	Born **14 March** in **Ulm, Germany**; family moves to **Munich**
Childhood	Spoke late (about age two and a half); playmates called him **"Brother Boring"**
Age 15	Leaves school in Munich, disliking its **regimentation**
1896	Studies at a university in **Zurich**, Switzerland; meets **Mileva Marić**
1902	Works as a **technical expert** at the **patent office in Bern**
1905	Publishes the **Special Theory of Relativity**; formula **$E = mc^2$**
1915	General Theory of Relativity
1919	Solar eclipse observations confirm his prediction; world fame
1921	**Nobel Prize in Physics**
1933	Emigrates to the **USA** when Hitler's Nazis come to power
1939	Letter to President **Roosevelt** warning about atomic bombs

Year	Event
1955	Dies on **18 April**, at age 76

Key points

- His mother once feared he was a "freak" because his head was large and he spoke late. He played the violin and loved music all his life.
- He was bright in school but hated rote learning and the rigid discipline of the Munich school; a teacher told him his presence spoiled the class.
- At university he thought for himself and often skipped lectures; Mileva shared his love of science. They married in 1903 but later divorced; he then married his cousin **Elsa**.
- The patent office job left him time to think, and in 1905 he published papers that changed physics.
- After the atom bomb destroyed **Hiroshima (1945)**, he was deeply shocked. He wrote to the **United Nations** proposing a **world government** and spent his last years campaigning for peace and democracy.
- *Time* magazine named him the **Person of the Century**.

WHY "BEAUTIFUL"?

Einstein's mind was brilliant in science and also compassionate: he used his fame to speak against war, nuclear weapons and injustice.

NOBEL FOR RELATIVITY?

His Nobel Prize (1921) was officially for explaining the **photoelectric effect**, not for relativity. The chapter mentions the prize; know the year.

Themes and values

- Independent thinking over rote learning.

- Genius needs freedom and curiosity.
- Science must serve humanity; responsibility of scientists.

LIKELY QUESTIONS

Why did Einstein leave school in Munich? Why did he write to Roosevelt, and why did he later regret it? How did he try to promote peace? Write a diary entry/letter as Einstein after Hiroshima.

Quick recap

- Ulm 1879, Munich, Zurich, Bern patent office.
- 1905 relativity and $E = mc^2$; Nobel 1921.
- Letter to Roosevelt 1939; after Hiroshima, letter to UN; died 1955; a pacifist.

Practice questions

1. Where was Albert Einstein born?
   (a) Munich (b) Ulm (c) Zurich (d) Bern
2. What nickname did his playmates give him?
   (a) Little Professor (b) Big Head (c) Silent Albert (d) Brother Boring
3. Why did Einstein leave his school in Munich?
   (a) He failed his exams (b) He disliked its regimentation and rote learning (c) His family moved to America (d) He fell ill
4. Where did Einstein work when he published his special theory of relativity?
   (a) At a university in Zurich (b) At the Nobel committee (c) At a school in Munich (d) At the patent office in Bern
5. In 1939 Einstein wrote a letter to President Roosevelt about
   (a) leaving Germany (b) the Nobel Prize (c) the danger of Germany building an atomic bomb (d) a world government
6. After the bombing of Hiroshima, Einstein
   (a) returned to Germany (b) stopped doing science (c) joined the army (d) wrote to the UN proposing a world government
7. Why did Einstein call himself, or why is he called, a pacifist? Give two incidents from the chapter.
8. "Einstein's life shows that the education system should encourage independent thinking." Discuss.

Answers: 1. b 2. d 3. b 4. d 5. c 6. d 7. see notes 8. see notes